

University of Prishtina “Hasan PRISHTINA”



Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026
Course: Data Preparation and Visualization

Week V – Data Preprocessing III

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What is today's agenda?

➤ **Today we are going to learn following things:**

- Data Reduction;
- Data Transformation;
- Data Discretization;
- Summary.

Data Reduction

➤ Data reduction:

- Representation of a reduced dataset that is much smaller in volume but produces the same (or nearly the same) analytical results;

➤ Why data reduction?

- A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete dataset.

➤ In a fact, what is data reduction?

- If the original data can be **reconstructed from the compressed data** without any **information loss**, the data reduction is called **lossless**.
- If, instead, we can reconstruct only an approximation of the original data, then the data reduction is called **lossy**.

Data Reduction: Strategies

➤ Data reduction strategies:

▪ Dimensionality reduction, e.g., remove unimportant attributes:

○ Wavelet transforms:

- The **discrete wavelet transform (DWT)** is a **linear signal processing** technique that, when applied to a data vector $\mathbf{X}$, transforms it to a numerically different vector, $\mathbf{X}'$, of wavelet coefficients. The two vectors are of the same length. When applying this technique to data reduction, we consider each tuple as an n -dimensional data vector, that is, $\mathbf{X} (x_1, x_2, \dots, x_n)$, depicting n measurements made on the tuple from n database attributes.

○ Principal Components Analysis (PCA):

- PCA **computes k orthonormal vectors** that provide a basis for the normalized input data. These are unit vectors that each point in a direction perpendicular to the others. These vectors are referred to as the principal components. The **input data are a linear combination** of the principal components.

○ Feature subset selection, feature creation:

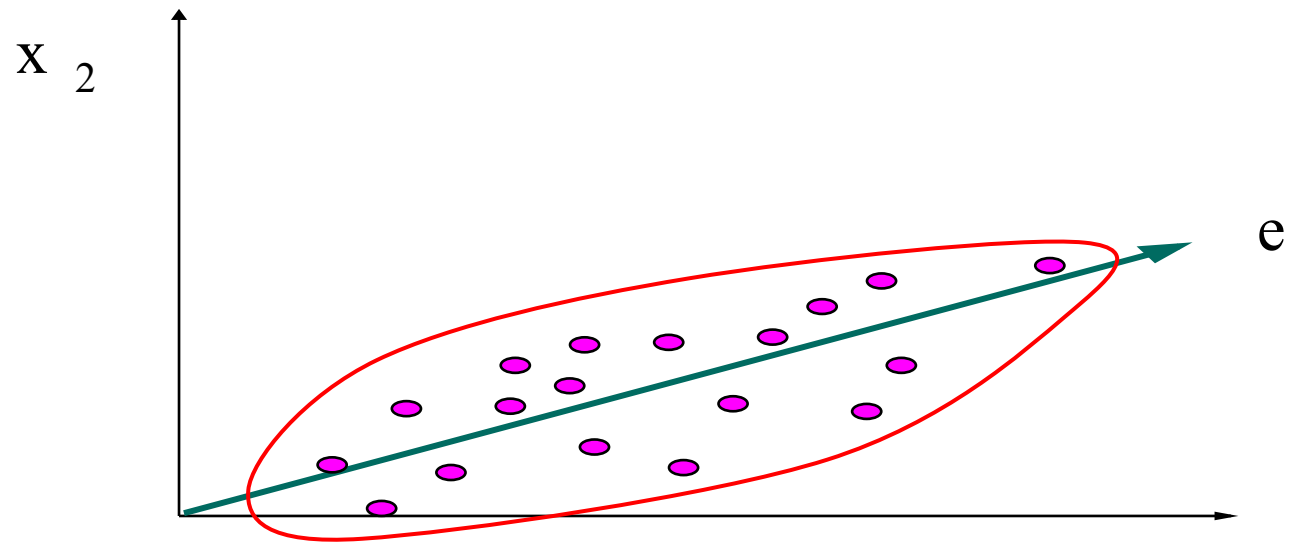
- Try to reduce the dataset size by removing **irrelevant** or **redundant attributes** (or dimensions). The **goal of attribute subset selection** is to **find a minimum set of attributes** such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes.

Data Reduction: Strategies

- Numerosity reduction (some simply call it: Data Reduction):
 - Regression and Log-Linear Models:
 - In (simple) **linear regression**, the data are modelled to fit a **straight line**. For example, a random variable, y (called a response variable), can be modelled as a linear function of another random variable, x (called a predictor variable), with the equation $y = \mathbf{w}x + \mathbf{b}$, where the variance of y is assumed to be constant. In the context of data mining, x and y are numeric database attributes.
 - Meanwhile, **Log-linear models** approximate discrete multidimensional probability distributions.
 - Histograms, clustering, sampling:
 - *Histograms* are highly effective at approximating both **sparse and dense data**, as well as highly skewed and uniform data;
 - *Clustering* is known as partition the objects into **groups, or clusters**, so that objects within a cluster are “**similar**” to one another and “**dissimilar**” to objects in other clusters;
 - *Sampling* can be used as a data reduction technique because it allows a **large dataset** to be represented by a **much smaller random data sample** (or subset).
 - Data cube aggregation.
- Data compression

Principal Component Analysis - PCA

- Find a projection that captures the largest amount of variation in data;
- The original data are projected onto a much smaller space, resulting in dimensionality reduction;
- We find the eigenvectors of the covariance matrix, and these eigenvectors define the new space.



Principal Component Analysis (Steps)

- Given N data vectors from n -dimensions, find $k \leq n$ orthogonal vectors (*principal components*) that can be best used to represent data:
 - Normalize input data: Each attribute falls within the same range;
 - Compute k orthonormal (unit) vectors, i.e., *principal components*;
 - Each input data (vector) is a linear combination of the k principal component vectors;
 - The principal components are sorted in order of decreasing “significance” or strength;
 - Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, it is possible to reconstruct a good approximation of the original data).
- Works for numeric data only.

Attribute Subset Selection

- Another way to reduce dimensionality of data:
- **Redundant attributes:**
 - Duplicate much or all the information contained in one or more other attributes;
 - E.g., purchase price of a product and the amount of sales tax paid.
- **Irrelevant attributes:**
 - Contain no information that is useful for preprocessing task at hand;
 - E.g., students' ID is often irrelevant to the task of predicting students' GPA.

Heuristic Search in Attribute Selection

- There are 2^d possible attribute combinations of d attributes;
- Typical **heuristic attribute** selection methods:
 - Best single attribute under the attribute independence assumption: **choose by significance tests**;
 - Best step-by-step feature selection:
 - The **best single-attribute** is picked first;
 - Then **next best attribute condition** to the first, ...
 - Step-by-step attribute elimination:
 - Repeatedly eliminate the worst attribute.
 - Best combined attribute selection and elimination;
 - Optimal branch and bound:
 - Use attribute elimination and backtracking.

Attribute Creation (Feature Generation)

- Create new attributes (features) that can capture the important information in a dataset more effectively than the original ones;
- **Three general methodologies:**
 - **Attribute extraction:**
 - Domain-specific.
 - **Mapping data to new space** (see: data reduction):
 - E.g., Fourier transformation, wavelet transformation, manifold approaches (not covered).
 - **Attribute construction:**
 - Combining features;
 - Data discretization.

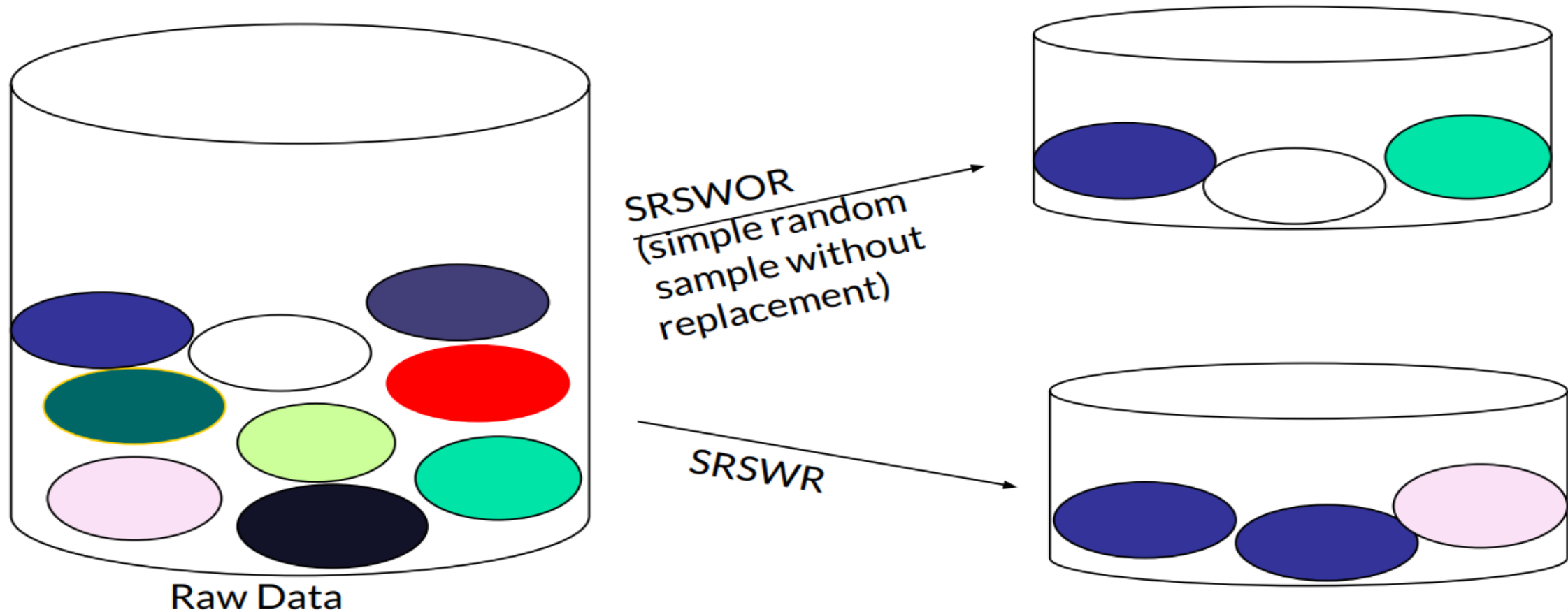
Data Reduction 2: Numerosity Reduction

- Reduce data volume by choosing alternative, smaller forms of data representation;
- Parametric methods (e.g., regression):
 - Let's suppose that the data fits some model, estimate **model parameters, store only the parameters, and discard the data (except possible outliers)**;
 - Ex.: Log-linear models - obtain value at a point in m-D space as the product on appropriate marginal subspaces.
- Non-parametric methods:
 - Do not assume models;
 - Major families: histograms, clustering, sampling ...

Sampling

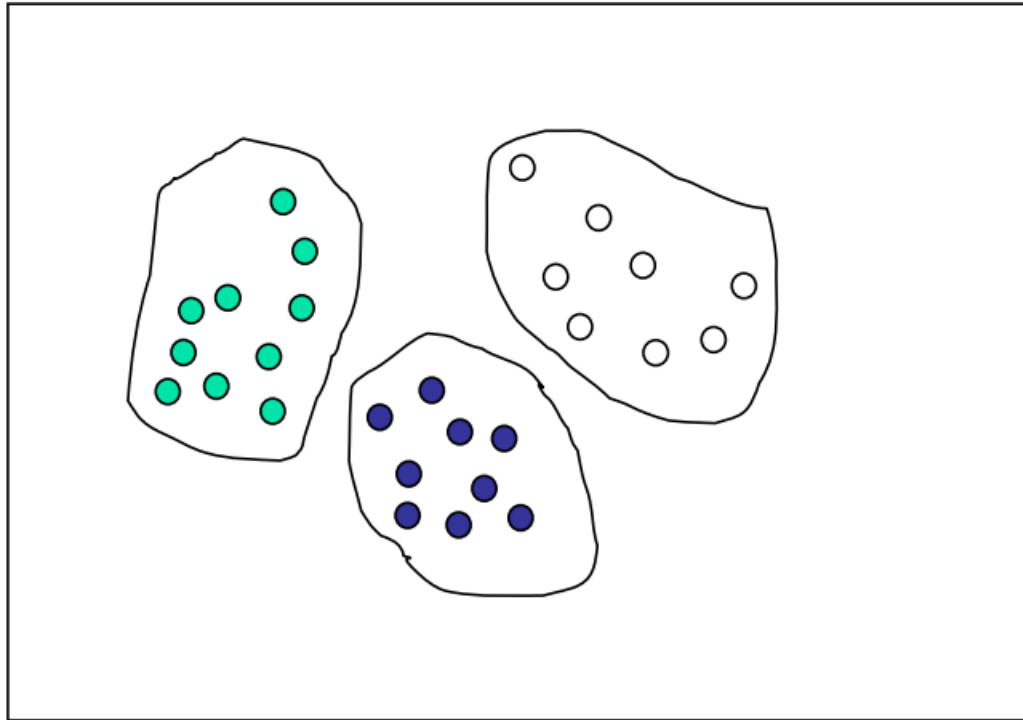
- Sampling: obtaining a small sample s to represent the whole dataset N ;
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data;
- Key principle: Choose a representative subset of the data:
 - Simple random sampling may have very poor performance in the presence of skew;
 - Develop adaptive sampling methods, e.g., stratified sampling.
- Note: Sampling may not reduce database I/Os (page at a time).

Sampling With or Without Replacement

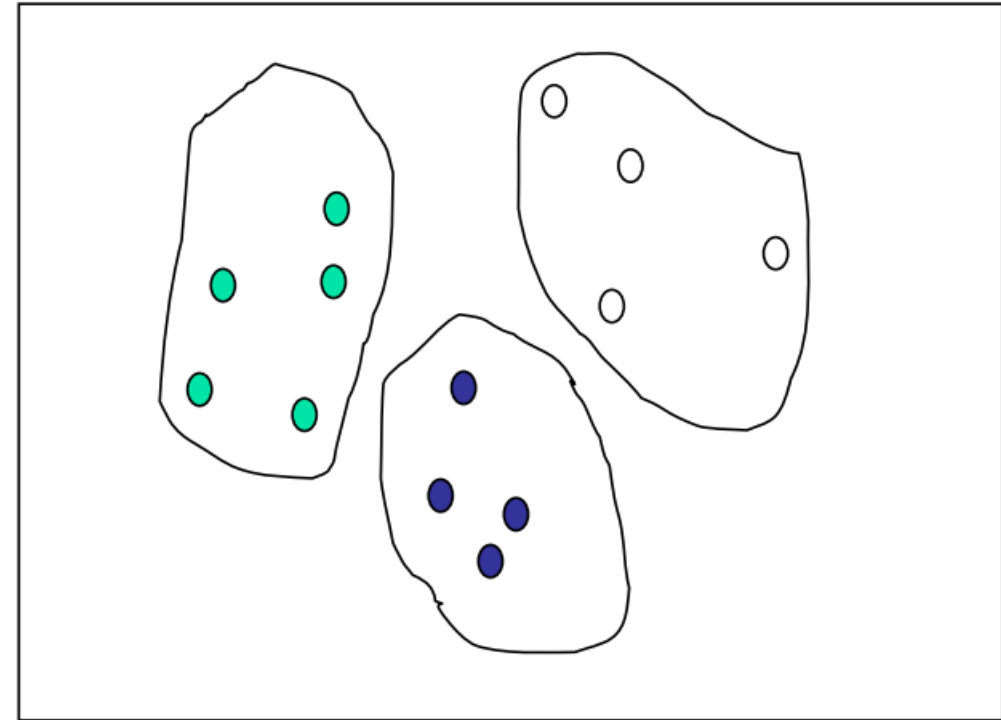


Sampling: Cluster or Stratified Sampling

Raw Data



Cluster/Stratified Sample



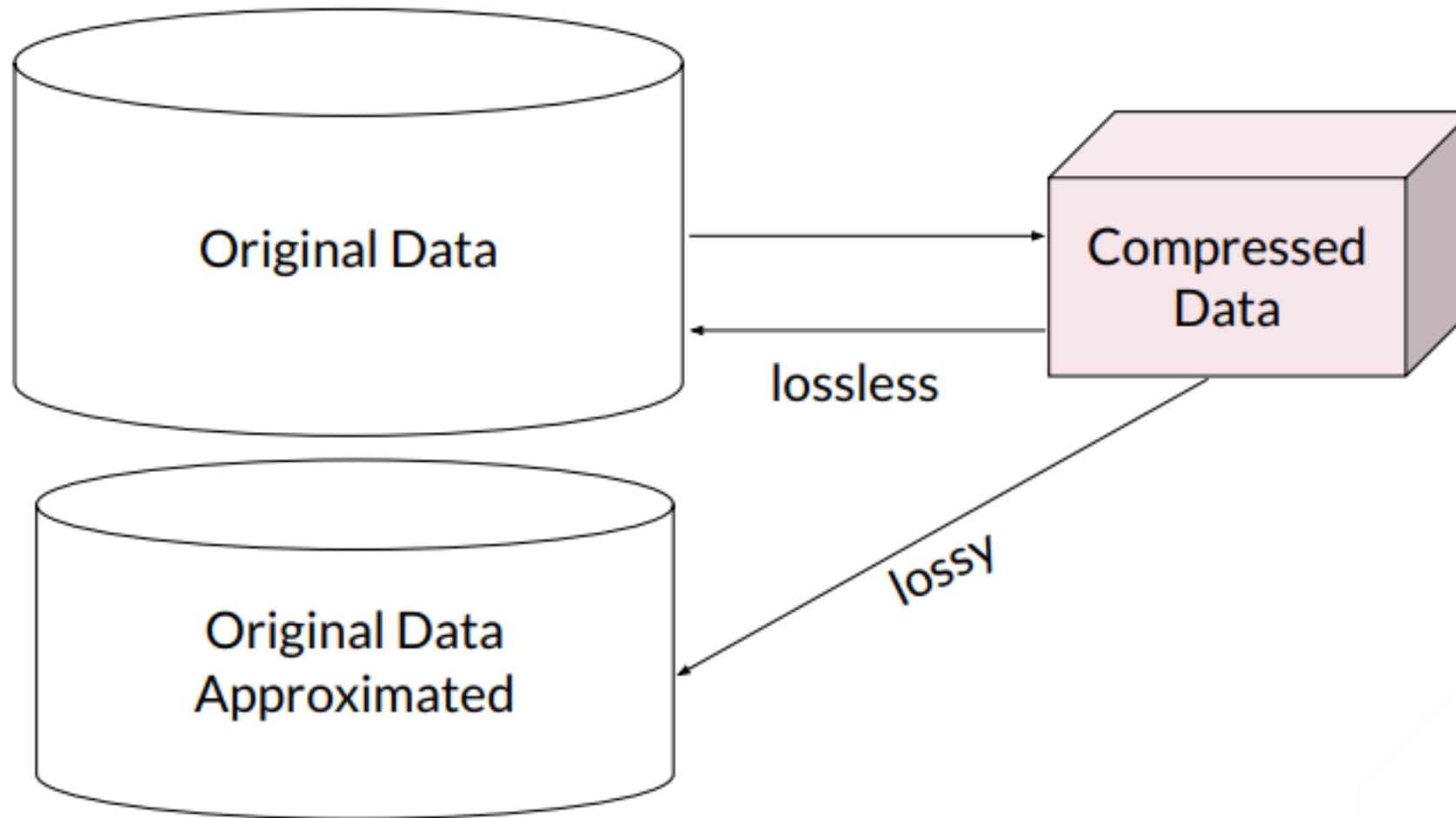
Data Cube Aggregation

- The lowest level of a data cube (base cuboid):
 - The aggregated data for an individual entity of interest;
 - E.g., a customer in a phone calling data warehouse.
- Multiple levels of aggregation in data cubes:
 - Further reduce the size of data to deal with.
- Reference appropriate levels:
 - Use the smallest representation which is enough to solve the task.
- Queries regarding aggregated information should be answered using data cube, when possible.

Data Reduction 3: Data Compression

- String compression:
 - There are extensive theories and well-tuned algorithms;
 - Typically lossless, but only limited manipulation is possible without expansion.
- Audio/video compression:
 - Typically, lossy compression, with progressive refinement;
 - Sometimes small fragments of signal can be reconstructed without reconstructing the whole.
- Time sequence is not audio:
 - Typically, short and vary slowly with time
- Dimensionality and numerosity reduction may also be considered as forms of data compression.

Data Compression



Data Transformation

- The measurement unit used can affect the data analysis. For example, changing measurement units from **meters** to **inches** for height, or from **kilograms** to **pounds** for weight, may lead to very different results;
- Methods:
 - Smoothing: Remove noise from data;
 - Attribute/feature construction:
 - New attributes constructed from the given ones.
 - Aggregation: Summarization, data cube construction;
 - Normalization: Scaled to fall within a smaller, specified range:
 - min-max normalization;
 - z-score normalization;
 - normalization by decimal scaling.
 - Discretization: Concept hierarchy climbing.

Normalization

- Normalizing the data attempts to give all attributes an equal weight;
- **Min-max normalization:** to $[new_min_A, new_max_A]$:
 - let A be a numeric attribute with n observed values, $v_1, v_2, \dots, v_n$.

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Ex. let income range \$12,000 to \$98,000 normalized to $[0.0, 1.0]$.
- Then \$73,600 is mapped to: $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

Normalization [cont.]

- **Z-score normalization/ zero-mean normalization:** (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

- E.x. Let $\mu = 54,000$, $\sigma = 16,000$. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

- **Normalization by decimal scaling:**

$$v' = \frac{v}{10^j} \quad \text{Where } j \text{ is the smallest integer such that } \text{Max}(|v'|) < 1$$

Discretization

➤ Three types of attributes:

- Nominal - values from an unordered set, e.g., color, profession;
- Ordinal - values from an ordered set, e.g., military or academic rank;
- Numeric - real numbers, e.g., integer or real numbers.

➤ **Discretization**: Divide the range of a continuous attribute into intervals:

- **Interval labels** can then be used to replace actual data values;
- **Reduce data** size by discretization;
- **Supervised** vs. **unsupervised**;
- **Split** (top-down) vs. **merge** (bottom-up);
- **Discretization** can be performed recursively on an attribute;
- Prepare for further analysis, e.g., **classification**.

Data Discretization Methods

- Typical methods: All the methods can be applied recursively:
 - **Binning:**
 - Top-down split, unsupervised.
 - **Histogram analysis:**
 - Top-down split, unsupervised.
 - **Clustering analysis** (unsupervised, top-down split or bottom-up merge);
 - **Decision-tree analysis** (supervised, top-down split);
 - **Correlation (e.g., χ^2) analysis** (unsupervised, bottom-up merge).

Simple Discretization: Binning

➤ Equal-width (**distance**) partitioning:

- Divides the range into N intervals of equal size: *uniform grid*;
- if A and B are the lowest and highest values of the attribute, the width of intervals will be $W = (B-A)/N$;
- The most straightforward, but outliers may *dominate presentation*;
- Skewed data is not handled well.

➤ Equal-depth (**frequency**) partitioning:

- Divides *the range into N intervals*, each containing approximately same number of samples;
- Good **data scaling**;
- *Managing categorical attributes* can be *tricky*.

Binning Method for Data Smoothing

- Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
 - Partition into equal-frequency (equi-depth) bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 26, 28, 29, 34
 - Smoothing by bin means:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 29, 29, 29, 29
 - Smoothing by bin boundaries:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 26, 26, 26, 34

Binning Method examples

- Here are data shown in string values: 8, 19, 16, 10, 22, 24, 6, 32, 17, 38, 59, 25?
- Partition into equal-frequency (equi-depth) bins?
 - Smoothing by bin means?
 - Smoothing by bin boundaries?

Binning Method examples

➤ Here are data shown in string values: 11, 19, 32, 5, 3, 24, 64, 8, 16?

- Partition into equal-frequency (equi-depth) bins?
- Smoothing by bin means?
- Smoothing by bin boundaries?

Binning Method examples

➤ Here are data shown in string values: 7, 9, 26, 14, 33, 42, 46, 2, 57, 17, 38?

- Partition into equal-frequency (equi-depth) bins?
- Smoothing by bin means?
- Smoothing by bin boundaries?

Concept Hierarchy Generation

- Concept hierarchy organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse;
- Concept hierarchies facilitate drilling and rolling in data warehouses to view data in multiple granularity;
- Concept hierarchy formation: Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for age) by higher level concepts (such as youth, adult, or senior);
- Concept hierarchy can be automatically formed for both numeric and nominal data. For numeric data, use discretization methods shown.

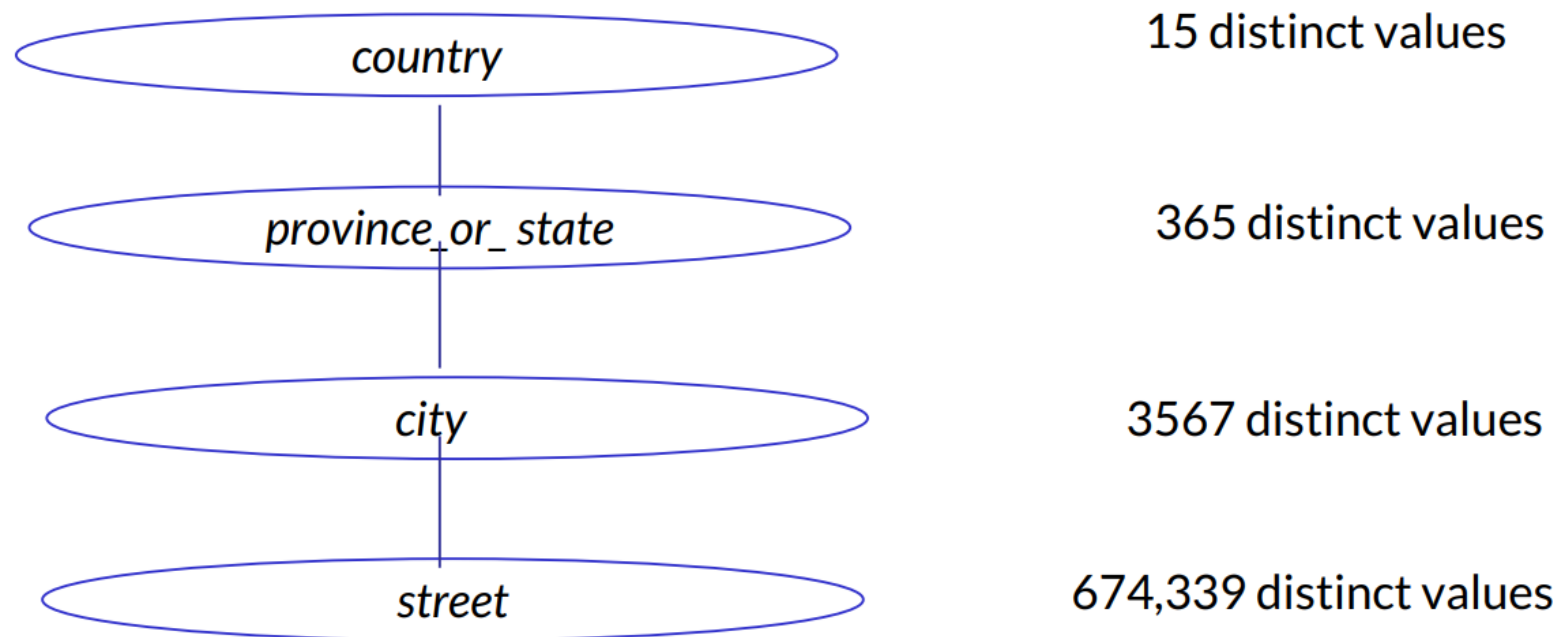
Concept Hierarchy Generation For Nominal Data



- Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts:
 - $\text{street} < \text{city} < \text{state} < \text{country}$.
- Specification of a hierarchy for a set of values by explicit data grouping:
 - $\{\text{Urbana, Champaign, Chicago}\} < \text{Illinois}$
- Specification of only a partial set of attributes:
 - E.g., only $\text{street} < \text{city}$, not others.
- Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values:
 - E.g., for a set of attributes: $\{\text{street, city, state, country}\}$.

Concept Hierarchy Generation

- Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
- The attribute with the most distinct values is placed at the lowest level of the hierarchy;
 - Exceptions, e.g., weekday, month, quarter, year.



Summary

- **Data quality:** accuracy, completeness, consistency, timeliness, believability, interpretability;
- **Data cleaning:** e.g. missing/noisy values, outliers;
- **Data integration from multiple sources:**
 - Entity identification problem;
 - Remove redundancies;
 - Detect inconsistencies.
- **Data reduction:**
 - Dimensionality reduction;
 - Numerosity reduction;
 - Data compression.
- **Data transformation and data discretization;**
- **Normalization:**
 - Concept hierarchy generation.

Questions?

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